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Iron-Limited Erythropoiesis in Dogs and Cats: A Systematic Review and Meta-Analysis of Current Evidence Examining Mean Corpuscular Volume, Mean Reticulocyte Volume, and Reticulocyte Hemoglobin Content

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ABSTRACT

Background: Iron-limited erythropoiesis (ILE) is a common condition in dogs and cats, which can lead to anemia; therefore, monitoring with erythrocyte and reticulocyte indices is recommended.

Objectives: To compare the values of mean corpuscular volume (MCV), mean reticulocyte volume (MCVr), and reticulocyte hemoglobin content (CHr) in dogs and cats with ILE.

Methods: Systemative review and meta-analysis. We conducted a systematic search using PRISMA criteria in PubMed, ScienceDirect, and Google Scholar up to 2024. It focused on erythrocyte and reticulocyte indices, such as MCV, MCVr, and CHr, in dogs and cats with ILE.

Results: This meta-analysis included eight articles. For dogs, the random effect sizes were 2.86 (0.55–5.18) for MCV, 2.18 (0.87–3.58) for MCVr, and 4.73 (1.37–8.08) for CHr. For cats, the effect sizes were 0.85 (0.19–1.5) for MCV, 3.45 (0.49–6.41) for MCVr, and 2.51 (0.29–4.74) for CHr. The analysis revealed I^2 values of 97%, 94.3%, and 98.2% in dogs, and 63.1%, 93%, and 95.1% in cats, for MCV, MCVr, and CHr, respectively. The overall random effects were 1.98 for MCV, 2.54 for MCVr, and 3.87 for CHr.

Conclusion and Clinical Importance: The findings revealed significant differences in reticulocyte indices, MCVr in cats, and CHr in dogs between the ILE-affected and the healthy groups. Considerable variability among studies indicates caution in generalizing findings and makes conclusions less definitive.

1 | Introduction

Iron is an essential element in the bodies of vertebrates that plays a fundamental role in the structure and function of hemoglobin [1–3]. Iron-limited erythropoiesis (ILE) results from a negative iron imbalance [1, 4]. In dogs and cats, ILE due to malnutrition and malabsorption occurs rarely, and the most common causes of ILE in animals are chronic bleeding,

Abbreviations: 95% CI, 95% confidence interval; CHCMr, mean corpuscular hemoglobin concentration of reticulocytes; CHr, reticulocyte hemoglobin content; ES, estimated; FeDef, Fe deficiency; ID, iron deficiency; IDA, iron deficiency anemia; ILE, iron-limited erythropoiesis; MCH, mean corpuscular hemoglobin; MCHC, mean corpuscular hemoglobin concentration; MCV, mean corpuscular volume; MCVr, mean reticulocyte volume; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RDW, red cell distribution width; RI, reference interval; SD, standard deviation.

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portosystemic shunt, inflammatory diseases, chronic kidney disease (CKD), and immune-mediated hemolytic anemia (IMHA) [5–7].

Traditional red blood cell (RBC) indices, such as mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), and mean corpuscular hemoglobin concentration (MCHC), mainly characterize mature erythrocytes [8–10]. Given the long lifespan of erythrocytes (120 days in humans and dogs and 70 days in cats), ILE needs to persist for weeks to months before changes in these indices become apparent. As a result, conventional RBC indices are relatively insensitive during the early stages of iron deficiency (ID), meaning that acute or subacute cases of ILE can often go unrecognized [1, 8, 11].

The reticulocyte indices commonly reported in companion animal literature include the reticulocyte hemoglobin content (CHr), mean reticulocyte volume (MCVr), and the mean corpuscular hemoglobin concentration of reticulocytes (CHCMr) [12]. The reference intervals (RIs) of these indices in canines have been well-documented for ADVIA hematology analyzers [1]. Some studies have demonstrated promising results and firmly suggested the reliability of these indices in identifying ILE [1]. However, other research has reported an inability of these reticulocyte indices to distinguish absolute from functional ILE and has found similar abnormalities present in dogs with other conditions [13].

Recently, reticulocyte indices have been used to reflect the functional iron available for erythropoiesis over the past 2 to 4 days [5, 8, 11]. Low CHr and MCVr levels are associated with hematologic and biochemical indicators of ILE [1, 5]. Additionally, reticulocyte indices are promising in detecting functional ILE, even before the development of overt microcytosis, which helps us detect early stages of ILE even before anemia [8, 10, 12]. Despite evidence showing that reticulocyte indices, particularly CHr, can identify early stages of ILE, neither CHr nor iron metabolism variables such as serum iron concentration and total iron-binding capacity (TIBC) can reliably distinguish between underlying diseases that ultimately lead to ILE [14]. This is due to overlapping results across different disease groups, the lack of a proper gold standard (such as iron staining or hemosiderin in cat bone marrow samples) [15], and the absence of other biochemical tests (such as ferritin

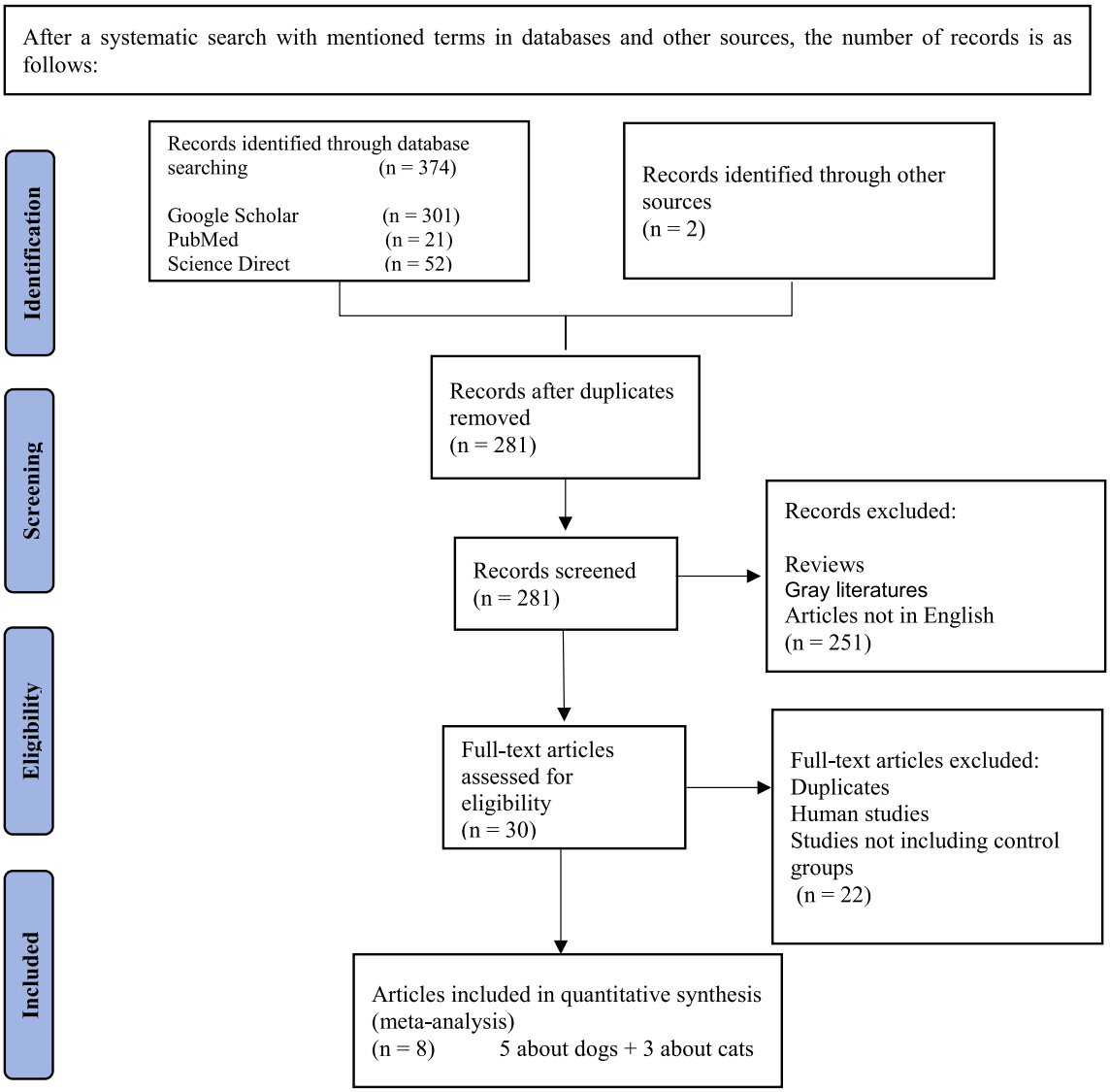

FIGURE 1 | Flow diagram for inclusion of studies in the combined systematic review and meta-analysis. *Source:* Moher et al. [25].

and soluble transferrin receptor) in veterinary medicine [5]. Early diagnosis of ILE might improve treatment efficiency by providing early opportunities for interventional procedures, thereby improving the effectiveness of treatment.

Considering the lack of systematic reviews regarding the evaluation of reticulocyte indices in dogs and cats affected by ILE, along with the differing objectives in previous studies, this study was conducted as a systematic review and meta-analysis aimed at examining the impact of ILE on the indices of MCV, MCVr, and CHr. We have thoroughly analyzed numerous articles and data to support our conclusion. We found multiple articles discussing the impact of ILE on erythrocyte and reticulocyte indices in dogs and cats, either narratively or statistically [5–8, 10, 12–14, 16–24], and finally, used those with enough information. In other words, the investigations have been summarized to demonstrate their value in characterizing ILE, and reticulocyte indices also receive attention in both veterinary and human medicine.

2 | Methods

2.1 | Systematic Review Protocol Development

The meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [25]. Although this review does not evaluate an intervention, the primary research question was structured using the PECO framework, which is appropriate for exposure-focused systematic reviews. Specifically, P (population) refers to dogs and cats, E (exposure) refers to the presence of ILE, c (comparison) refers to healthy controls, and O (outcome) refers to changes in erythrocyte and reticulocyte indices (MCV, MCVr, and CHr). The primary question of this review was: Can erythrocyte and reticulocyte indices, such as MCV, MCVr, and CHr, be utilized in monitoring ILE in dogs and cats, based on the current evidence? The research question was raised and revised in several stages, and initial investigations were conducted in existing scientific resources. Given the use of previously published data, this study did not require ethical approval.

TABLE 1 | Selection criteria used to include or exclude articles.

Inclusion criteria	Exclusion criteria
English language articles	Non-English language articles
Original/research articles	Review articles and Human studies
Articles with numerical data	Articles with no numerical data or gray literatures
Studies with control groups	Studies not including control groups
Studies focusing on ID or ILE and MCV, MCVr, CHr in dogs and cats	Studies focusing solely on ID or ILE in dogs and cats

Abbreviations: CHr, reticulocyte hemoglobin content; ID, iron deficiency; MCVr, reticulocyte mean cell volume.

TABLE 2 | Characteristics of included studies: Species, analytical devices, case and control groups, and evaluated variables.

Authors	Year	Country	Species	Device (for hematology analyses)	Case group/ condition	Case group/ number	Control group/ condition	Control group/ number	Researched variables
Fry and Kirk	2006	United States	Dog	ADVIA 120	Nutritional ID	7	Healthy	7	CHr, MCVr, MCV
Radakovich et al.	2015	United States	Dog	ADVIA 120	Low CHr	47	Normal CHr	49	CHr, MCVr, MCV
Schaefer and Stokol	2015	United States	Dog	ADVIA 2120	FeDef	11	Healthy	122	CHr, MCVr, MCV
Melendez-Lazo et al.	2015	Spain	Dog	ADVIA 120	Non-regenerative Anemia	24	Healthy	16	CHr, MCVr
Foy et al.	2015	United States	Dog	ADVIA 120	Blood Donors	13	Non-donors	20	CHr, MCVr, MCV
Hunt and Jugan	2021	United States	Cat	ADVIA 2120i	Decreased Serum Iron	7	Normal Serum Iron	13	CHr, MCVr, MCV
Keiner et al.	2020	Germany	Cat	ADVIA 2120	ILE	20	Non-ILE	225	CHr, MCV
Betting et al.	2022	Switzerland	Cat	ADVIA 2120i	ID	9	Non-ID	55	CHr, MCVr, MCV

Abbreviations: CHr, reticulocyte hemoglobin content; FeDef, Fe deficiency; ID, iron deficiency; ILE, iron-limited erythropoiesis; MCV, mean corpuscular volume; MCVr, reticulocyte mean cell volume.

2.2 | Search Strategy

Two authors (Y.H. and P.A.) independently reviewed the articles. Both authors deemed the articles suitable for inclusion in the meta-analysis, and the study included only articles that were considered appropriate by both authors. Articles deemed suitable by only one author were referred to a third person (MA) for the final decision, and their opinion was used as the criterion. An extensive literature review evaluated the potential of MCV, MCVr, and CHr biomarkers for characterizing ID, iron deficiency anemia (IDA), or ILE in dogs and cats using PubMed, ScienceDirect, and Google Scholar. The search incorporated various terms, including "Iron deficiency" OR "Iron deficiency anemia" OR "Iron-limited erythropoiesis" AND dog OR cat OR "canine" OR "feline" AND "Mean corpuscular volume" OR "Mean reticulocyte volume" OR "Reticulocyte hemoglobin content." Full-text research articles that described the clinical characteristics of patients with ID, IDA, or ILE and applied MCV, MCVr, and CHr as diagnostic variables were considered eligible for meta-analysis. English-language studies published between 2004 and 2024 were considered. No relevant systematic reviews on this topic were found. Prospective/retrospective cohort, case–control, and cross-sectional studies were deemed suitable for the meta-analysis (Figure 1).

2.3 | Study Selection

In the initial screening phase, titles and abstracts were examined to identify potentially relevant studies, followed by a more detailed evaluation of the selected studies. The selection of studies was carried out based on specific inclusion and exclusion criteria. Additionally, duplicate records, review articles, case reports, expert opinions, letters, and editorials were excluded from the review process. Additionally, studies that included only human subjects or used MCV, MCVr, and CHr as biomarkers unrelated to ID, IDA, and ILE were excluded. Articles without accessible numerical data (gray literature), those published in languages other than English, and studies focusing solely on ID, IDA, and ILE were removed from consideration (Table 1). Given the study's objective of examining changes in the MCVr

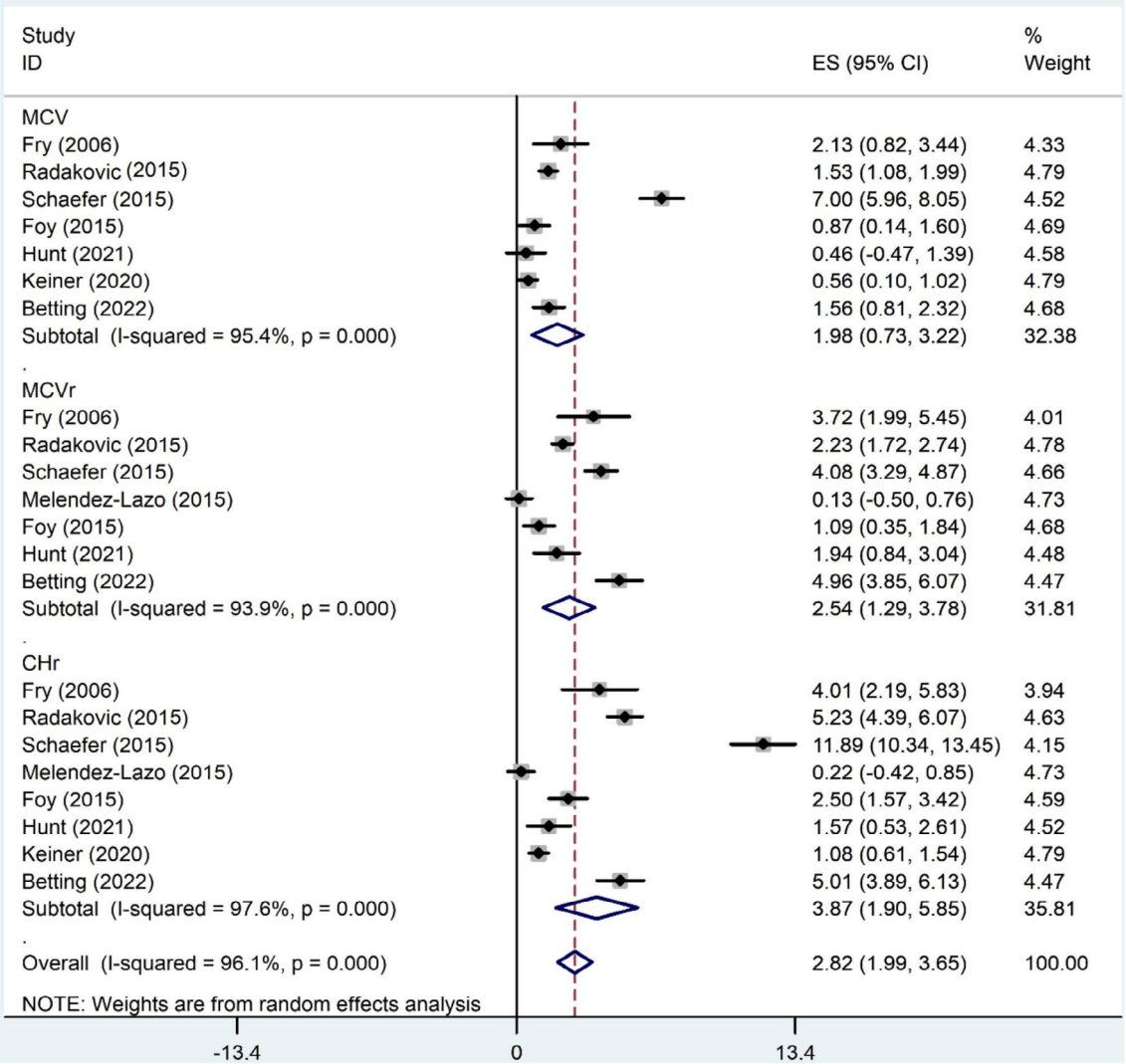

FIGURE 2 | Forest plot for individual and pooled estimated (ES) amounts of effect size for the difference between the two groups with and without anemia. The vertical line shows the value of zero, and the vertical dashed line shows the value of the pooled estimate. The squares show the point estimates, and the lateral lines show the 95% confidence intervals for the estimate of each study. The diamond shows the pooled estimates' point and 95% confidence interval. *Note:* Weights are from random effects analysis.

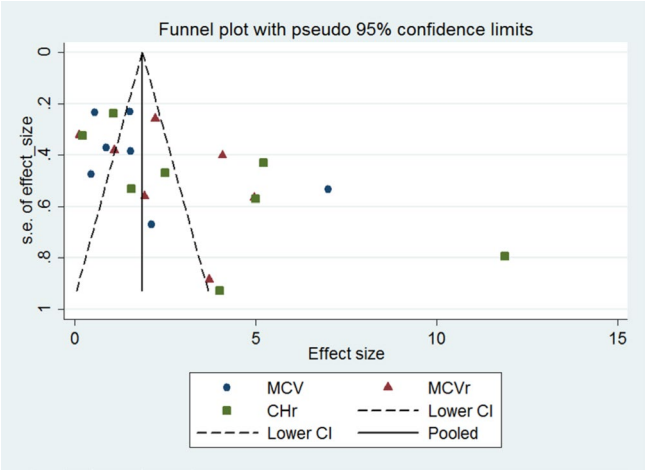

FIGURE 3 | The funnel plot of MCV, MCVr, and CHr for the graphical assessment of publication bias in the meta-analysis with pseudo 95% confidence limits. A symmetric inverted funnel shape arises from a "well-behaved" dataset, in which publication bias is unlikely. As can be seen in this figure, this symmetry cannot be attributed to the graph, mainly due to the relatively small number of retrievable articles.

and CHr indices in dogs and cats affected by ILE, observational studies (cohort, case–control, and cross-sectional) were selected for inclusion in the review. These study designs provide appropriate comparative data between animals with ILE (as the exposure group) and healthy animals (the control group), making them more suitable for assessing these indices in animals affected by ILE. Since the aim was not to investigate the effect of a therapeutic intervention, clinical trials and interventional studies were excluded from the analysis. In total, the eight included studies comprised 316 dogs and 329 cats, providing a combined sample size of 645 animals. The smallest sample size was reported in Fry and Kirk [16] with only 7 cases, while the largest belonged to Keiner et al. [8] with 245 cases.

2.4 | Data Extraction

The following information was systematically collected: details of the articles (including the first author and publication year), the country of study, the species involved, sex, specifications of the devices used, the status of case/control participants, and the variables investigated (Table 2). Data extraction was conducted independently and in parallel by two researchers (Y.H. and P.A.) to minimize the likelihood of errors and bias. A list of articles excluded in the eligibility stage and the reason for each removal is present in Data S1.

2.5 | Data Analysis

To enable a comparison of results from different studies, which recorded different measurement scales, and to limit any possible influence posed by measuring small sample sizes, all effect sizes were transformed into Hedges' *g*. It is assumed that this standardized index is a modification of Cohen's *d*, an unbiased measure of effect size achieved by applying a correction factor; this is of great advantage for small sample sizes. This transformation enables us to combine the results of various studies and present them in a manner that facilitates interpretation and understanding. Effect size measures tell us how much one group differs from another. There are many methods to calculate it; one of the most well-known is Cohen's *d* method. Cohen's *d* is the difference between two means divided by the standard deviation of the data. Cohen's *d* and Hedges' *g* are remarkably similar measures of effect size, except when sample sizes are below 20; in this case, Hedges' *g* is a corrected effect size. For sample sizes greater than 20, the results for both statistics are roughly equivalent. Funnel and forest plots are the most popular graphical representations for meta-analysis results. Funnel plots provide a proper graphical representation of the presence of bias, and forest plots represent the heterogeneity of findings within studies included in a meta-analysis. A meta-analysis was performed in Stata 14, using the "meta" command to pool the effect size estimates with the random effects method. A forest plot was generated for each analysis. Publication bias was assessed using a funnel plot plotted against the data, and heterogeneity was assessed by the I-squared index, in which values greater than 50 were considered heterogeneous. However, due to data limitations, a systematic risk of bias assessment was not performed in individual studies using standardized tools such as the Newcastle–Ottawa Scale, which constitutes a limitation of this meta-analysis. Data obtained from dogs and cats were separately included in subgroup analyses. To better examine and control the heterogeneity resulting from the combination of different species (dog and cat) and various variables (MCV, MCVr, CHr), we created separate forest plots for each species and each variable. This disaggregated analysis is essentially a subgroup analysis of the species and variables, allowing us to investigate the specific effects of each group separately and avoid the influence of heterogeneous combined activity.

3 | Results

3.1 | Assessment of MCV, MCVr, and CHr in Dogs and Cats With ILE

Using a random-effects model, the I^2 values for MCV, MCVr, and CHr were 95.4%, 93.9%, and 97.6%, respectively, in dogs and cats (Figure 2), indicating high heterogeneity among the included articles ($p < 0.001$). We used a funnel plot with pseudo 95% confidence limits and a scatterplot to visualize the effect size of interest against the measure of study precision, represented by the standard error of the effect size. The funnel plot is used primarily as a visual aid for detecting publication bias. An asymmetric inverted funnel shape arises from a "well-behaved" dataset in which publication bias is unlikely. Figure 3 shows that this symmetry cannot be attributed to the graph, mainly due to the relatively small number of retrievable articles. The random effect sizes for MCV, MCVr, and CHr are 1.98, 2.54, and 3.87, respectively. The 95% confidence intervals (CIs) for MCV vary between 0.73 and 3.22, MCVr between 1.29 and 3.78, and CHr between 1.90 and 5.85, as shown in Table 3.

3.2 | Assessment of MCV, MCVr, and CHr in Dogs With ILE

After thoroughly reviewing multiple sources, we identified five studies that provide relevant data on the dog's MCV, MCVr, and CHr

TABLE 3 | CHr, MCVr, and MCV: mean ± SD, effect size, and 95% CI in included studies.

	CHr			MCVr			MCV		
Included studies	**Mean ± SD**	**Effect size**	**95% CI**	**Mean ± SD**	**Effect size**	**95% CI**	**Mean ± SD**	**Effect size**	**95% CI**
Fry (2006, dog)									
ID ($n = 7$)	19.50 ± 2.20	4.01	2.19–5.83	71.30 ± 5.30	3.72	1.99–5.45	65.50 ± 2.30	2.13	0.82–3.44
Control ($n = 7$)	25.80 ± 0.30			86.40 ± 2.20			69.90 ± 1.80		
Radakovic (2015, dog)									
ID ($n = 47$)	21.25 ± 0.80	5.23	4.38–6.07	79.50 ± 5.16	2.23	1.72–2.74	63.00 ± 2.80	1.53	1.07–1.99
Control ($n = 49$)	25.20 ± 0.71			92.00 ± 6.00			68.00 ± 3.65		
Schaefer (2015, dog)									
ID ($n = 11$)	16.82 ± 1.05	11.89	10.33–13.45	73.45 ± 5.33	4.08	3.29–4.87	58.40 ± 5.40	7.00	5.96–8.04
Control ($n = 122$)	26.70 ± 0.81			88.30 ± 3.46			69.60 ± 0.60		
Melendez-Lazo (2015, dog)									
ID ($n = 24$)	24.50 ± 3.40	0.22	−0.41 to 0.85	84.80 ± 7.55	0.13	−0.50 to 0.76	—	—	—
Control ($n = 16$)	25.10 ± 1.18			83.90 ± 5.62			—		
Foy (2015, dog)									
ID ($n = 13$)	24.60 ± 0.58	2.49	1.57–3.41	88.80 ± 2.01	1.09	0.34–1.83	67.40 ± 1.03	0.87	0.14–1.60
Control ($n = 20$)	25.90 ± 0.48			90.80 ± 1.71			68.60 ± 1.56		
Total (random effects)									
ID ($n = 102$)		4.73	1.37–8.08		2.18	0.87–3.58		2.86	0.55–5.18
Control ($n = 214$)									
Hunt (2021, cat)									
ID ($n = 7$)	16.70 ± 0.83	1.56	0.53–2.60	55.00 ± 4.16	1.94	0.84–3.03	45.00 ± 2.50	0.46	−0.47 to 1.39
Control ($n = 13$)	18.00 ± 0.83			62.00 ± 3.30			46.00 ± 2.00		
Keiner (2020, Cat)									
ID ($n = 20$)	14.69 ± 1.57	1.07	0.61–1.54	—	—	—	39.80 ± 3.08	0.56	0.10–1.02
Control ($n = 225$)	16.49 ± 1.68			—			43.45 ± 6.70		
Betting (2022, cat)									
ID ($n = 9$)	15.60 ± 0.53	5.01	3.89–6.12	51.40 ± 1.73	4.96	3.85–6.07	42.10 ± 0.71	1.56	0.80–2.31
Control ($n = 55$)	17.70 ± 0.40			57.00 ± 1.01			43.10 ± 0.63		
Total (random effects)									
ID ($n = 36$)		2.51	0.29–4.74		3.45	0.49–6.41		0.85	0.19–1.51
Control ($n = 293$)									
Overall (random effects)									
ID ($n = 138$)		3.87	1.90–5.85		2.54	1.29–3.78		1.98	0.73–3.22
Control ($n = 507$)									

Abbreviations: 95% CI, 95% confidence interval; CHr, reticulocyte hemoglobin content; ID, iron deficiency; MCV, mean corpuscular volume; MCVr, reticulocyte mean cell volume; SD, standard deviation.

in the presence of ILE. These studies were included in the meta-analysis. Using a random effect model, I-squared values of 97% (Figure 4), 94.3% (Figure 5), and 98.2% (Figure 6) were obtained for MCV, MCVr, and CHr, respectively, indicating high heterogeneity among the included articles ($p < 0.001$). The random effect sizes for MCV, MCVr, and CHr are 2.86, 2.18, and 4.73, respectively. The 95% CIs for MCV vary between 0.55 and 5.18, MCVr between 0.87 and 3.58, and CHr between 1.37 and 8.08, as shown in Table 3.

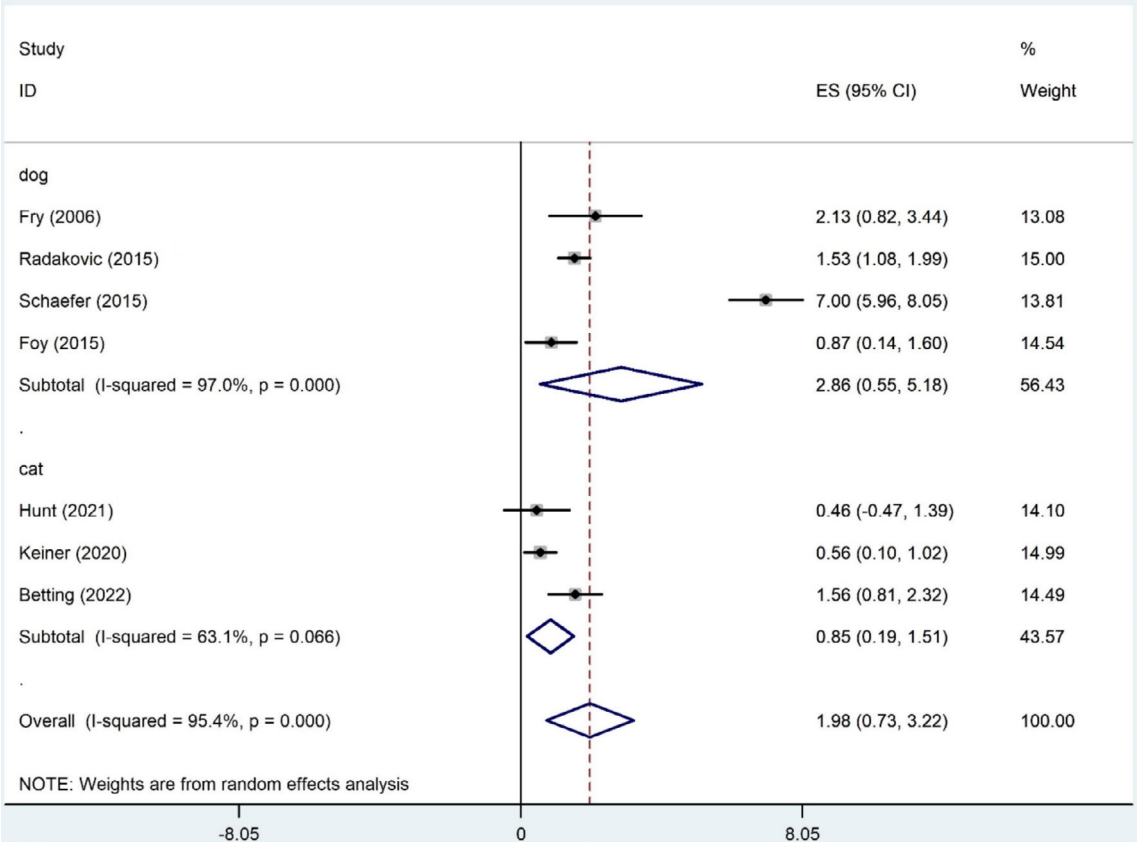

FIGURE 4 | Forest plot for individual and pooled ES amounts of effect size for difference of MCV between the two groups with and without anemia in cats and dogs separately. The vertical line shows the value of zero, and the vertical dashed line shows the value of the pooled estimate. The squares show the point estimates, and the lateral lines show the 95% confidence intervals for the estimate of each study. The diamond shows the pooled estimates' point and 95% confidence interval. *Note:* Weights are from random effects analysis.

3.3 | Assessment of MCV, MCVr, and CHr in Cats With ILE

Our meta-analysis included three studies that evaluated the cat's MCV, MCVr, and CHr in relation to ILE, which were selected following a thorough review of the relevant literature. Using a random-effects model, the I^2 values for MCV, MCVr, and CHr were 63.1% (Figure 4), 93% (Figure 5), and 95.1% (Figure 6), respectively, indicating high heterogeneity among the included articles ($p < 0.001$). The random effect sizes for MCV, MCVr, and CHr are 0.85, 3.45, and 2.51, respectively. The 95% CIs for MCV vary between 0.19 and 1.5, MCVr between 0.49 and 6.41 (Figure 5) for MCVr, and CHr between 0.29 and 4.74, as shown in Table 3.

4 | Discussion

At the outset, we will summarize the methods employed in each article included in this study to explain the reasoning for its incorporation into the statistical analysis. In Fry 2006, the researchers induced nutritional ID in healthy dogs [16]. They did this by feeding them an iron-deficient diet for 35 days. The Radakovic 2015 article discussed subtracting absolute ID from functional ID [13]. It measured how inflammatory factors relate to CHr in dogs. These factors include white blood cells and acute-phase proteins. The Schaefer 2015 study examined the usefulness of reticulocyte indices [7]. They can tell IDA apart from three other diseases in dogs. It used healthy dogs to set RIs. They selected the dogs based on a brief history and physical exam. The intervals are for important blood indices. Then, according to these RIs and many other criteria, we identified the dogs by their ID. In Melendez-Lazo's 2015 study, he investigated the effect of inflammation on reticulocyte indices [17]. He found that inflammation causes functional ILE. A 2015 article studied the effect of blood donation on ID [18]. It used reticulocyte indices in dogs.

Dogs donating blood six or more times yearly will show signs of ID based on their reticulocyte indices. However, these changes might not appear in their biochemistry or mature RBC indices. Hunt's 2021 study aimed to determine the prevalence of ILE in cats with gastrointestinal disease [19]. The Keiner 2020 article established RIs for CHr and RETIC-HGB in healthy cats. It then studied the usefulness of these two indices [8]. They diagnose ILE. The 2022 Betting study aims to investigate the use of reticulocyte indices [20]. They diagnose CKD and chronic hematuria. Subcutaneous ureteral bypasses relate to these.

ILE is one of the important disorders in dogs and cats, with considerable implications in small animal medicine. Furthermore, ILE can cause various problems in the body before it leads to anemia; therefore, early diagnosis and

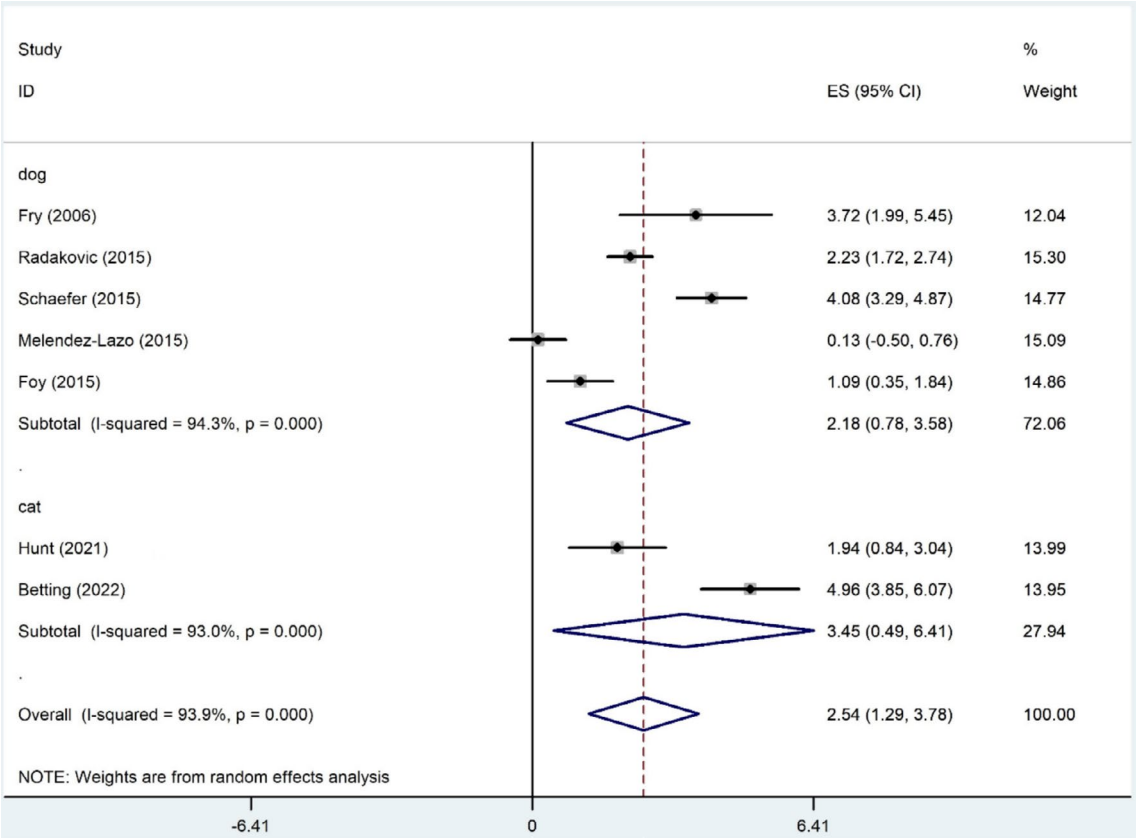

FIGURE 5 | Forest plot for individual and pooled ES amounts of effect size for the difference of MCVr between the two groups with and without anemia in cats and dogs separately. The vertical line shows the value of zero, and the vertical dashed line shows the value of the pooled estimate. The squares show the point estimates, and the lateral lines show the 95% confidence intervals for the estimate of each study. The diamond shows the pooled estimates' point and 95% confidence interval. *Note:* Weights are from random effects analysis.

intervention in the management of this disease are of paramount importance. This deficiency occurs when tissue stores of iron deplete [26, 27], a condition that can be managed with proper nutrition and medical intervention. The reasons are insufficient dietary intake of iron, iron malabsorption, blood loss, portosystemic shunt, and any factor affecting ferritin function, such as inflammation [28, 29], which are practical issues veterinarians encounter daily.

The diagnosis of ILE involves both direct and indirect methods. Direct methods typically analyze blood serum to measure levels of key analytes, including iron, ferritin, and transferrin. Conversely, indirect methods evaluate the alterations in RBC characteristics that result from a decrease in iron levels. These characteristics include MCH, MCHC, MCV, and red cell distribution width (RDW) [27, 30, 31].

Since ILE appears in old diagnostic tests over time, reticulocyte indices have gained prominence in recent years. These can identify ILE in the earliest possible stages and prevent its consequences. The most notable reticulocyte indices, in order of preference, are CHr, MCVr, and CHCMr. Based on this prioritization, we selected articles that included CHr [12, 32]. In seven out of eight articles, MCVr was also present, but CHCMr needed to be more available.

Objective criteria revealed that the case groups in eight selected articles exhibited varying degrees of ILE. By comparing the MCV, MCVr, and CHr of these groups with those of the control groups, we found significant differences between the case and control groups. Nevertheless, further actions were taken to prove the effectiveness of these variables in characterizing ILE.

As expected in this study, MCV, MCVr, and CHr have been reduced in those groups with confirmed ILE; however, the reduction in CHr was more significant than in the other variables. Afterward, the meta-analysis revealed that the effect size of CHr in all dog groups was more substantial than that of MCVr and MCV, as expected. Due to the magnitude of the MCVr effect size compared to other indices, when using CHr as a variable to characterize ILE, it is essential to proceed with caution. The higher MCVr effect size might be due to the following reasons: insufficient studies were available for cats, and one of the articles related to cats used in this study lacked MCVr data. We also consider MCVr a more reliable variable for characterizing ILE in cats.

Although the overall results of the combined meta-analysis were presented, subgroup analyses were conducted for dogs and cats, as well as for each variable separately, to control for heterogeneity arising from different species and variables. The results are presented as forest plots, which facilitate a better understanding of the specific effects of species and variables. However, given the biological differences between species and the distinct nature of the measured variables, caution is needed when interpreting the results. Despite the limited number of studies, the I^2 statistic is a powerful

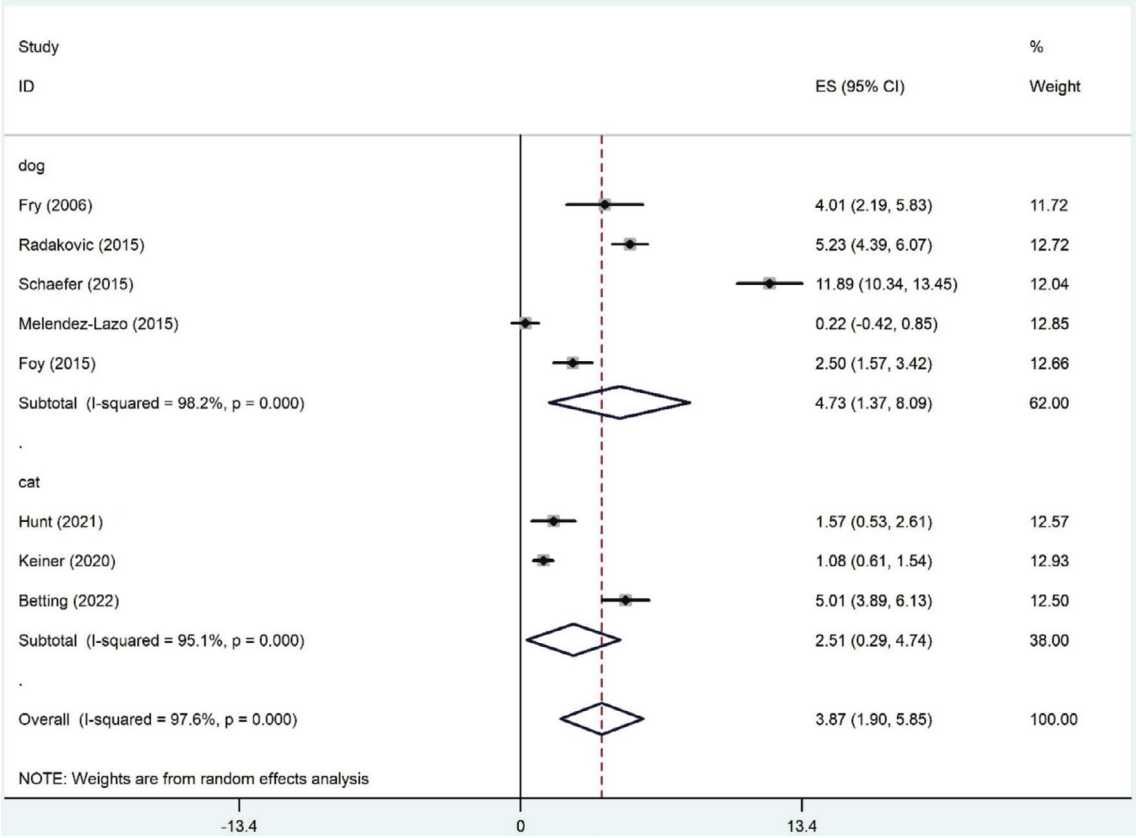

FIGURE 6 | Forest plot for individual and pooled ES amounts of effect size for difference of CHr between the two groups with and without anemia in cats and dogs separately. The vertical line shows the value of zero, and the vertical dashed line shows the value of the pooled estimate. The squares show the point estimates, and the lateral lines show the 95% confidence intervals for the estimate of each study. The diamond shows the pooled estimates' point and 95% confidence interval. *Note:* Weights are from random effects analysis.

and accepted tool for assessing and estimating the degree of heterogeneity. This enhanced the analytical power and data interpretation in the study. However, the main limitation of I-squared is its inability to identify the causes of heterogeneity.

5 | Limitations

One major limitation of this meta-analysis is the absence of a comprehensive risk of bias assessment in the included primary studies. The only tool used was the funnel plot for evaluating publication bias; a systematic risk of bias evaluation using standardized tools, such as the Newcastle–Ottawa Scale, was not performed. Furthermore, the limited number of studies and the high heterogeneity observed (I^2 values ranging from 63% to 98%) increase the likelihood of selection and reporting biases. Therefore, the results should be interpreted with caution, and it is recommended that future research conduct more systematic evaluations of the quality and bias of the included studies to enhance the robustness of the results obtained from the meta-analysis.

6 | Conclusion

The results of this meta-analysis indicate significant differences in these indices between groups with ILE and healthy control groups. Specifically, CHr in dogs and MCVr in cats showed the most substantial differences, which might help in monitoring this condition. Important variability among studies indicates caution in generalizing findings and makes conclusions less definitive. The current results support the view that reticulocyte indices can be valuable tools for early diagnosis and monitoring of ILE in veterinary medicine, especially in situations where traditional RBC indices have not yet changed.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Authors declare no institutional animal care and use committee or other approval was needed. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.